

mxd - an R package for MaxDiff in R

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Introduction

xxx

Motivation

xxx

Statement of need

xxx

Workflow

In the first step, we load the `mxd` package and will have a look at the design how it is usually exported.

```
library(mxd)
design <- readRDS("../data/design.rds")
head(design)
```

```
##   ID Set Position Item Response
## 1  1  1         1    10       -1
## 2  1  1         2     3        0
## 3  1  1         3     9        0
## 4  1  1         4     4        1
## 5  1  2         1    11        0
## 6  1  2         2     6        1
```

To convert this into the right format, we use the `csv_to_dm()` function. For the `anchor` argument (default set to `NULL`), the user can choose from both `direct` or `indirect`. For the `type` of coding, the user can decide between `best-worst`, `best-worst-seq` (first best, afterwards worst from a reduced choice set), `worst-best-seq`, `best-only` (only considering best choices), `worst-only`, `maxdiff` (maximum difference), and `exploded` (exploded logit paired comparison). For our example, we stick to the industry standard.

```
dm <- csv_to_dm(
  design = design,
  id = ID,
  cs = Set,
  item = Item,
  ch = Response,
  anchor = "direct",
  mxd_tasks = 16L,
  type = "best-worst"
)
```

```
head(dm)
```

```
##   ID Set alt Item_1 Item_2 Item_3 Item_4 Item_5 Item_6 Item_7 Item_8 Item_9
## 1  1  1  1     0     0     1     0     0     0     0     0     0
## 2  1  1  2     0     0     0     1     0     0     0     0     0
## 3  1  1  3     0     0     0     0     0     0     0     0     0
## 4  1  1  4     0     0     0     0     0     0     0     0     1
## 5  1  2  1     0     0    -1     0     0     0     0     0     0
## 6  1  2  2     0     0     0    -1     0     0     0     0     0
##   Item_10 Item_11 Item_12 Item_13 Item_14 Item_15 Item_16 Item_17 choice
## 1         0         0         0         0         0         0         0         0
## 2         0         0         0         0         0         0         0         1
## 3         1         0         0         0         0         0         0         0
## 4         0         0         0         0         0         0         0         0
## 5         0         0         0         0         0         0         0         0
## 6         0         0         0         0         0         0         0         0
```